

- `inc ecx`
- `cmp ecx, n`
- `jl for`
- `popa`
- `leave`
- `ret`

The above code is insertion sort implemented in the NASM dialect of assembly. The same code implemented for a different architecture, such as ARM or MIPS would look completely different. Attribution: http://dgit.in/NASM_Wiki_Attr

Tools and Learning Resources

As we have made it abundantly clear, assembly language is highly influenced by the target platform on which it will run. So the best way to get information about assembly language straight from the source, the CPU designers themselves.

Whatever hardware platform you are trying to target, just search for the CPU architecture followed by ‘instruction set manual’, and you’re likely to find detailed PDF manuals with everything you’d need to know. For example, search for ‘ARM instruction set manual’ or ‘Intel instruction set manual’.

While these manuals are great, they are mostly targeted towards people who already have some knowledge of assembly. Getting into Assembly language is tough, and require a different approach than other languages.

Firstly you will need to learn about machine architectures, and then the actual code. It can be hard to figure all this out on your own. Luckily there are many websites available nowadays that offer college course lecture videos, projects, and assignments online and for free. One such great course is titled “The



At 1500 pages Intel's Software Developer's Manuals are the deepest look you can get at their processor architecture and instruction set.